Shravani Mahadeshwar

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Education

Frankfurt School of Finance and Management — Frankfurt
Master of Applied Data Science — Sep 2025 - Sep 2027 (Expected)
Grade: 2.0 (German scale, 1.0 highest)
Relevant Coursework: Machine Learning, Deep Learning, Databases and Cloud Computing

University of Mumbai — Mumbai
Bachelor of Science in Data Science and Business Analytics — Sep 2021 - Mar 2024
Grade: 9.52 / 10.00
Relevant Coursework: Time Series Analysis, Natural Language Processing, Computer Vision

Skills

Programming: Python, R, SQL, Bash
Machine Learning: Scikit-learn, XGBoost, TensorFlow, PyTorch, Statistical Modeling, Feature Engineering
Deep Learning & NLP: Neural Networks, Transformer Architectures, RAG, Sentiment Analysis, Text Classification
Explainability: SHAP, LIME, Integrated Gradients
Data Engineering: PySpark, Databricks, MLflow, AWS S3, AWS EC2
Deployment & Visualization: Docker, Flask, Git, Linux, Streamlit, Power BI
Languages: English (C1), German (B1)

Professional Experience

JM Financial Services Limited — Mumbai
Junior Data & Business Analyst — Jun 2024 - Jan 2025

- Designed and maintained **4+ enterprise Power BI dashboards** integrating multi-source financial databases to surface client yield metrics and KPIs for executive stakeholders.
- Architected a **production-ready OCR and signature validation pipeline** using **OpenCV, EasyOCR, Flask APIs, and Dockerized deployment**, reducing KYC manual screening by **60%**.
- Engineered a **Python-Outlook automation framework** for recipient-specific report distribution with validation and tracking logic, cutting reporting turnaround time by **70%**.
- Applied **RFM analysis and K-Means clustering** with targeted feature engineering to support high-yield customer segmentation strategies.

JM Financial Services Limited — Mumbai
Data Science Intern — Sep 2023 - Feb 2024

- Automated mutual fund report processing through a **Selenium-based RPA workflow** integrated with FTP systems, reducing manual operational effort by **90%**.
- Designed an **NLP pipeline for NIFTY 500 financial news** using **VADER sentiment analysis**, transforming unstructured textual data into structured daily investment signals and trend indicators.

Projects

LLM Audit Bench — Mar 2026

- Built a modular trustworthiness auditing pipeline evaluating **5 open-source LLMs** (367M–1.6B params) across fairness, robustness, explainability, privacy, and transparency using **CrowS-Pairs stereotype benchmarking, SHAP-based Gini attribution scoring, and PII generation risk detection**, revealing no model exceeded the **random fairness baseline** on stereotype preference, with a **6x robustness variance** across models.

Interpreting Transformer Hallucinations — Feb 2026

- Investigated how GPT-Neo-125M generates factual hallucinations by mapping token-level attribution across **144 attention heads**, identifying a **sparse 3-node circuit** responsible for incorrect outputs and demonstrating that grammatical structure drives **52% of causal attribution** over semantically relevant tokens.

Certifications & Scholarships

- Deutschlandstipendium Scholar — Sep 2025
- McKinsey Forward Program — May 2025 – Jul 2025